Encrypting volumes with NetApp Volume Encryption

Cloud Manager 3.7

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January 03, 2020

This PDF was generated from https://docs.netapp.com/us-en/occm37/task_encrypting_volumes.html on December 03, 2020. Always check docs.netapp.com for the latest.

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NetApp Volume Encryption (NVE) is a software-based technology for encrypting data at rest one volume at a time. Data, Snapshot copies, and metadata are encrypted. Access to the data is given by a unique XTS-AES-256 key, one per volume.

About this task

- Starting with Cloud Manager 3.7.1, a NetApp Volume Encryption license is automatically installed on each Cloud Volumes ONTAP system that is registered with NetApp Support.
  - Adding NetApp Support Site accounts to Cloud Manager
  - Registering pay-as-you-go systems

  Cloud Manager does not install the NVE license on systems that reside in the China region.

- At this time, Cloud Volumes ONTAP supports NetApp Volume Encryption with an external key management server. An Onboard Key Manager is not supported.
- You need to set up NetApp Volume Encryption from the ONTAP CLI.

  You can then use either the CLI or System Manager to enable encryption on specific volumes. Cloud Manager does not support NetApp Volume Encryption from its user interface and from its APIs.

Learn more about supported encryption technologies.

Steps

1. Review the list of supported key managers in the NetApp Interoperability Matrix Tool.

   Search for the **Key Managers** solution.

2. Connect to the Cloud Volumes ONTAP CLI.
3. Install SSL certificates and connect to the external key management servers.

   ONTAP 9 NetApp Encryption Power Guide: Configuring external key management

4. Create a new encrypted volume or convert an existing unencrypted volume using either the CLI or System Manager.
   - CLI:
     - For new volumes, use the **volume create** command with the -encrypt parameter.

ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on a new volume

- For existing volumes, use the **volume encryption conversion start** command.

  ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on an existing volume with the volume encryption conversion start command

- System Manager:
  - For new volumes, click **Storage** > **Volumes** > **Create** > **Create FlexVol** and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Creating FlexVol volumes

  - For existing volumes, select the volume, click **Edit**, and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Editing volume properties

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